

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (CL) Examination

November 2012

CL402 Computer Aided Structural Analysis

Date: 05.12.2012, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Define: (i) primary nodes, (ii) secondary nodes. [02]
- (b) Do as directed: [05]
- (i) Evaluate: $\int L_1^2 L_2 L_3^4 L_4^3 dV$.
 - (ii) Define: shape function.
 - (iii) Write stress strain relationship for an-isotropic material.
 - (iv) Which are the types of elements are considered in FEM.
 - (v) Write a member flexibility matrix for a space truss.
- Q - 2 (a) Find the member end actions of the members of truss shown in figure: 1. The axial rigidity for the vertical and horizontal member is EA & for the diagonal member is 2EA. [09]
- (b) For iso-parametric elements shown in figure: 2, determine the local coordinates of the point Q which has Cartesian coordinates (8, 6) [05]

OR

- Q - 2 (a) Analyze the two span continuous beam as shown in figure: 3. Determine end actions, support reactions and joint displacement for the beam. [09]
- Take F_M matrix = $L/(12EI) [2L^2 \ 3L \ 0 \ 0; 3L \ 6 \ 0 \ 0; 0 \ 0 \ 4L^2 \ 6L; 0 \ 0 \ 6L \ 12]$
- B_{MS} matrix = $1/L [-1 \ 0 \ 1 \ -1; L \ 0 \ 0 \ L; 0 \ -1 \ 0 \ 1; 0 \ L \ 0 \ 0]$
- B_{RS} matrix = $1/L [1 \ 0 \ 1 \ 1; -1 \ 1 \ -1 \ -2; 0 \ -1 \ 0 \ 1]$
- (b) For the element shown in figure: 4 assemble Jacobian matrix corresponding to Gauss point (0.2254, 0.2254). [05]
- Q - 3 (a) Derive the B_{MS} and B_{RS} matrix for the frame as shown in figure: 5. Take EI = constant. [06]
- (b) (i) Write a short note on plane strain problem with neat sketch. [04]
- (ii) Determine shape functions for a two noded bar element with natural coordinate system as shown in figure: 6. Also draw the variation on it. [04]

OR

- Q - 3 (a) Analyze the frame shown in figure: 7 and derive the B_{MS} and B_{RS} matrix. Take EI = constant. Considering the joint displacements are the rotation at A, B and at C. [06]
- (b) (i) Write a short note on plane stress problem with neat sketch. [04]
- (ii) Using Lagrange function derive shape functions for the four noded rectangular element as shown in figure: 8. [04]

SECTION – II

Q - 4 (a) Analyze the propped cantilever beam loaded with UDL of 20 kN/m over the entire length by flexibility method system approach. The length of a beam is 5 m & $EI =$ constant. [04]

(b) Determine the stiffness matrix for a beam shown in figure: 9. $EI =$ constant. [03]

Q - 5 (a) Explain Rotation of Axes in Two Dimensions. [05]

(b) Using stiffness method member approach find unknown joint displacement and reaction for the truss shown in figure: 10 if, support A moves left by 2 mm. Take $AE = 32000$ kN. [09]

OR

Q - 5 (a) Analyse the continuous beam shown in figure: 11 using stiffness method member approach. Assume that supports are unyielding and EI as constant for all members. [14]

Q - 6 (a) Find unknown reaction for plane frame shown in figure: 12 using stiffness method member approach. Take $E = 2 \times 10^8$ kN/m², $I = 2 \times 10^{-3}$ m⁴ & $A = 0.03$ m². [14]

OR

Q - 6 (a) Briefly describe Symmetry and Anti – symmetry. [05]

(b) Find S_{FF} & A_{FC} for the grid shown in figure: 13. Take $GJ = 0.5 EI$ $E = 2 \times 10^5$ N/mm² & $I = 2 \times 10^{-3}$ m⁴. [09]

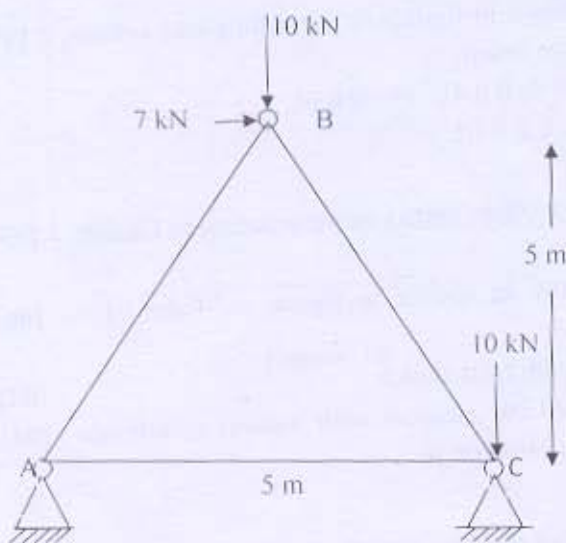


Figure: 1

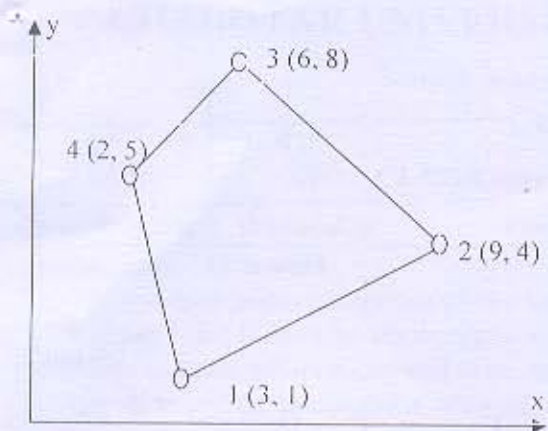


Figure: 2

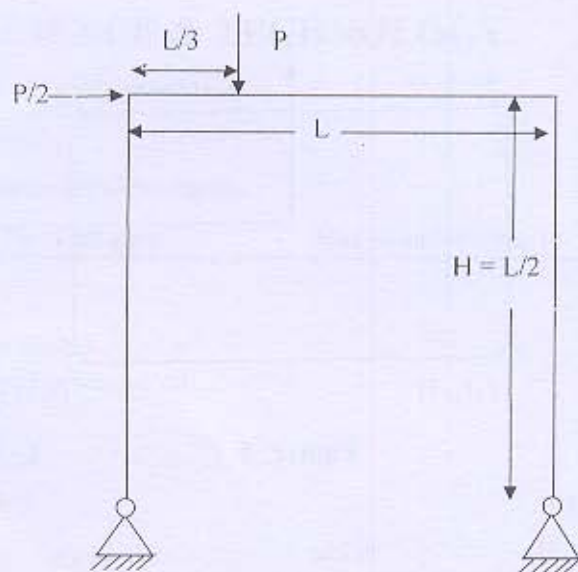


Figure: 5

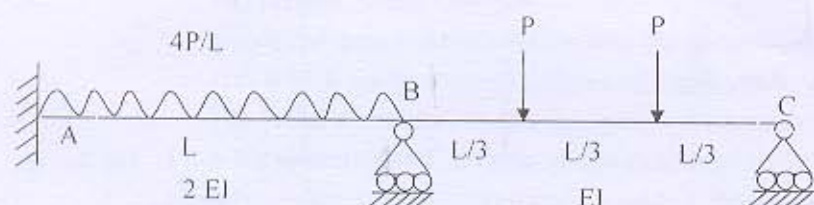


Figure: 3

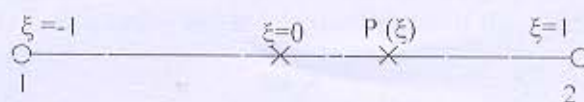


Figure: 6

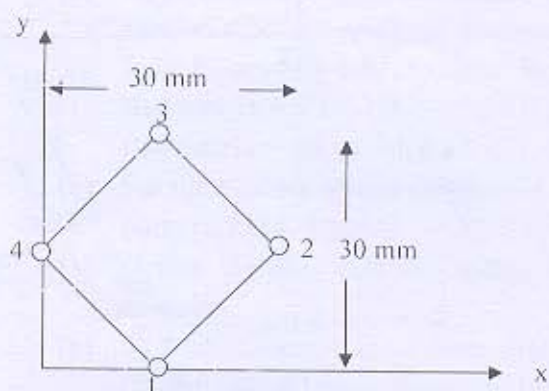


Figure: 4

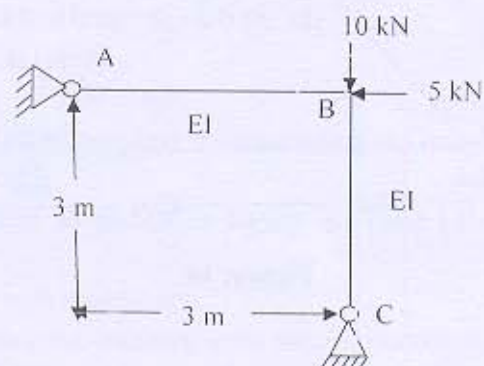


Figure: 7

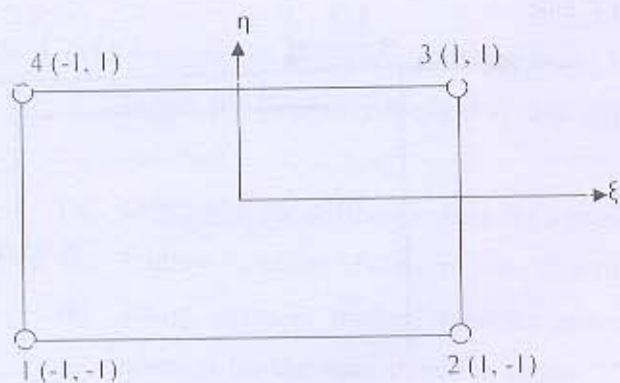


Figure: 8

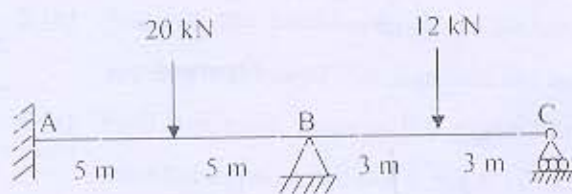


Figure: 9

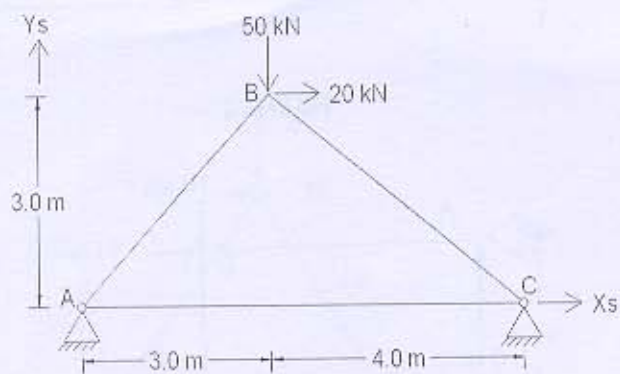


Figure: 10

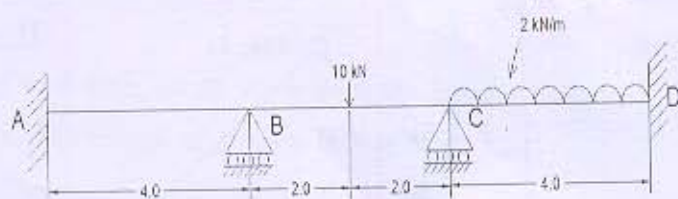


Figure: 11

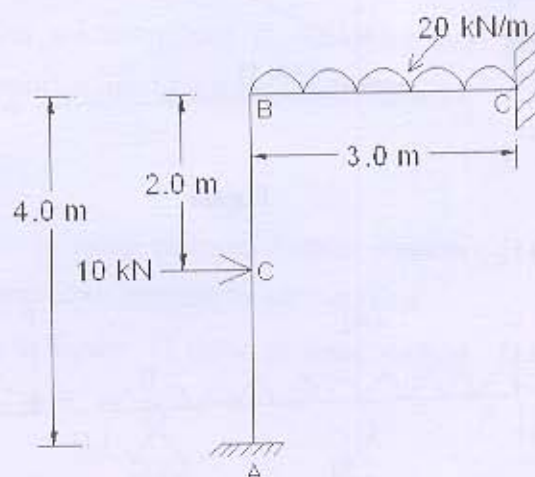


Figure: 12

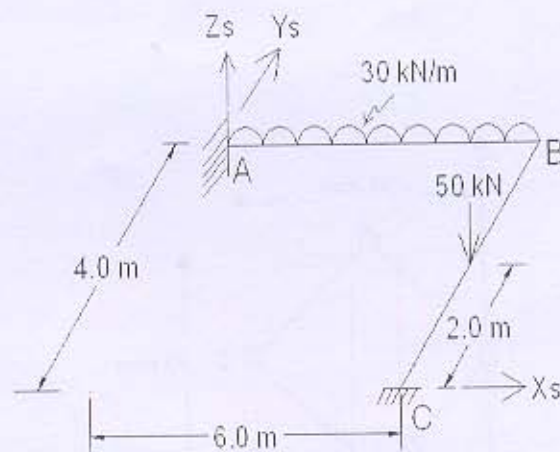


Figure: 13
